

A note on the method of calculating digestibility coefficients of nutrients provided by single ingredients to feeds of aquatic animals

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Abstract

A commonly used equation for the calculation of the apparent digestibility coefficients of feedstuffs for fish and crustaceans is shown to be incorrect. The correct (but rarely used) equation is presented and contrasted with the previous equation. Data from a previously published trial is used to illustrate the differences between these equations. In many cases, the values obtained using the two equations is very similar. Larger differences are found, however, when the level and digestibility of the nutrient in the test ingredient are greatly different from those in the reference diet.

KEY WORDS: digestibility, protein quality, methodology

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Introduction

Of the methods currently employed to determine the apparent digestibility coefficient of nutrients for feedstuffs, the most common involves comparison of the nutrient in the diet with that recovered in the faeces. When studying aquatic species, however, difficulties arise in quantifying the faecal material resulting from ingested feed and it is therefore necessary to measure the nutrient in the feed and faeces relative to an indigestible marker, typically chromic oxide (Furukawa & Tsukahara 1966). Since nutrient sources alone are generally not complete diets, the source under investigation is often combined with a reference diet (Kleiber 1961; Cho & Slinger 1979). The apparent nutrient digestibility of the test ingredient can then be calculated from a comparison of the apparent nutrient digestibility of the reference diet alone and in combination with the test ingredient. Two equations have been used for this calculation. One is described by Cho & Slinger (1979) and the other was described by Kleiber (1961). This short communication

contrasts these two equations, and discusses the inappropriateness of the first equation and the validity of the second.

The equation described by Cho & Slinger (1979) is:

$$\text{ADCN}_{\text{ingr}} = [\text{ADCN}_{\text{com}} - (100 - i) \times \text{ADCN}_{\text{ref}}]i^{-1} \quad (1)$$

where: ADCN = apparent digestibility coefficient of a nutrient; ingr = ingredient under investigation; ref = reference diet; com = diet comprised of a combination of reference diet and test ingredient; i = level of test ingredient in combined diet (%); (100 – i) = level of reference diet in combined diet (%).

Equation (1) has been used by the majority of authors using this method to calculate the nutrient digestibility of specific feedstuffs in aquatic animals (Cho & Slinger 1979; Hardy *et al.* 1984; Law 1984; de la Noüe & Choubert 1985; Wilson & Poe 1985; Brown *et al.* 1989; Reigh *et al.* 1990; Hajen *et al.* 1993; Sullivan & Reigh 1995; Aksnes *et al.* 1996; McGoogan & Reigh 1996; Appleford & Anderson 1997). The equation assumes that the nutrient digestibility of the combined diet is the average of the nutrient digestibility of the reference diet and the test ingredient, weighted by the proportion of each in the combined diet (100 – i and i, respectively). Clearly, unless the test ingredient and the reference diet contain equal levels of the nutrient under consideration, each component of the combined diet (i.e. the test ingredient and reference diet) will contribute differently to the level of the nutrient in the combined diet. Equation (1) does not, therefore, account for the relative contribution of the nutrient from the reference diet and the test ingredient to the combined diet, and is inappropriate for use in estimating nutrient digestibility. Instead, an equation that accounts for the relative nutrient contribution of the reference diet and the test ingredient to the combined diet should be used.

Such a method was described by Kleiber (1961; see also Schürch 1969 and Pfeffer *et al.* 1995) and is represented by

the following equation, which has been arranged for contrast with Equation (1):

$$\text{ADCN}_{\text{ingr}} = [(a + b) \times \text{ADCN}_{\text{com}} - (a) \times \text{ADCN}_{\text{ref}}]b^{-1}; \quad (2)$$

where: a = nutrient contribution of reference diet to nutrient content of combined diet, $= (\text{level of nutrient in reference diet}) \times (100 - i)$; b = nutrient contribution of test ingredient to nutrient content of combined diet, $= (\text{level of nutrient in test ingredient}) \times i$; i = level of test ingredient in combined diet (%); and $(a + b)$ = level of nutrient in combined diet (%).

As was the case for Equation (1), Equation (2) assumes that the nutrient digestibility of the combined diet is the weighted mean of the nutrient digestibilities of its components (i.e. reference diet and test ingredient). In Equation (1), however, the weighting is based on the relative proportions of the reference diet and the test ingredient, whereas in Equation (2), the weighting is based on the nutrient contribution of each of these components (i.e. by 'a' and 'b', respectively). The weighting in Equation (2) reflects the fact that test ingredients with higher or lower levels of the nutrient will have concomitantly greater or lesser influence on the calculated nutrient digestibility of the combined diet.

The extent to which the values for the digestibility coefficients obtained from Equations (1) and (2) differ depends upon the nutrient content and digestibility of the test ingredient. The value 'i' in Equation (1) is not related to the nutrient content of the test ingredient, whereas the corresponding variable, 'b' in Equation (2) is directly proportional to its nutrient content.

Example

To illustrate the extent of the differences in the values obtained from Equations (1) and (2), the data from a trial conducted by de la Noüe & Choubert (1985) to determine the digestibility coefficients of protein and energy from various species of invertebrates in rainbow trout were subjected to both equations. Four diets were used in this trial: a fishmeal-based reference diet and three experimental diets incorporating invertebrate biomass at the level of 300 g kg^{-1} in replacement of the reference diet. de la Noüe & Choubert (1985) used Equation (1) to calculate the apparent digestibility coefficients of nutrients and energy in the individual ingredients (Table 1). In this short communication, Equation (2) was also applied to the data provided and the resulting digestibility values for protein and energy were compared with those obtained in the original work.

Discussion

Although in most cases the differences in the values obtained from either Equation (1) or (2) were comparatively small, the difference in protein digestibility values for gammarid protein was considerable. The difference in values obtained from the two equations is greatest when the digestibility coefficient of the combined diet is greatly different from that of the reference diet and when the difference in the protein or energy level between the reference diet and the test ingredient is high.

For the past several years, there has been a great deal of interest in identification of alternative protein sources for use in feeds for aquatic animals. Many of the protein sources under consideration for use as partial replacements of fish

Table 1 Apparent digestibility coefficients for protein and energy of some invertebrate biomasses in rainbow trout¹

Component	Level ²	a ³	b ³	ADCe1 ⁴	ADCe2 ⁵	Difference
Protein (%)						
Chironomids	49.5	46.7	14.9	83.7	80.3	-3.4
Daphnids	56.2	46.7	16.9	82.7	79.8	-2.9
Gammarid	48.4	46.7	14.5	76.0	67.9	-8.1
Energy (kJ g ⁻¹)						
Chironomids	17.3	13.9	5.2	73.7	72.1	-1.6
Daphnids	20.4	13.9	6.1	80.7	80.9	0.2
Gammarid	17.2	13.9	5.1	65.4	62.7	-2.7

¹ Data from de la Noüe & Choubert (1985).

² Level of protein or energy in each test ingredient.

³ a = contribution of protein or energy to combined diet from reference diet; b = contribution of protein or energy to combined diet from test ingredient ($a + b$ = the level of the protein or energy in the combined diet (%)).

⁴ Apparent digestibility coefficient from Equation (1).

⁵ Apparent digestibility coefficient from Equation (2).

protein, especially those of plant origin, have considerably lower levels of digestible protein than do protein sources of animal origin, and so the values obtained from Equation (1) could be seriously in error. In any case, the values obtained from this equation do not reflect the differences in nutrient or energy contribution of the reference diet and the test ingredient to the combined diet and, therefore, are in error. The values obtained from Equation (2), however, do reflect the relative contribution of the nutrient and energy to the combined diet, and therefore this revised equation should be used instead of Equation (1). Accordingly, the results of the many digestibility studies that have been obtained using Equation (1) should be reexamined using Equation (2).

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